

(عدد الصفحات)الحجم المتوقع للنظام

PRD => 15-25
Use Cases => 10-20
FRS => 20-30
SRS => 30-50
System Design => 15-25
API Design => 20-40
Database Design => 10-20
Security Design => 10-15
Frontend Design => 15-25

Phase 1

Product Vision
PRD
User Personas
User Journey
Use Cases

Phase 2

FRS
Functional Modules
Business Rules
Acceptance Criteria

Phase 3

SRS
Non Functional Requirements
Constraints
Assumptions

Phase 4

System Design
Database Design
API Design
Security Design
Frontend Design

المراحل بالتفصيل

Phase 1 :

1. الصفحات المطلوبة (Required Pages)

بناءً على السكوب (Scope)، دي القائمة الأساسية للصفحات اللي هتحتاج تبنيها في الـ Frontend:

- مجموعة صفحات الهوية والتحقق (Auth Pages):
 - صفحة التسجيل (Register) والـ Login.
 - صفحة استعادة كلمة المرور (Forgot/Reset Password).
 - صفحة تأكيد الإيميل (Email Verification).
- الصفحات الرئيسية (Core Pages):
 - الرئيسية (Home/Feed): لعرض المنشورات (Posts) والقصص (Stories) الخاصة بالأشخاص اللي المستخدم متابعهم.
 - الملف الشخصي (Profile Page): وله شكلين: بروفايلك الشخصي (مع إمكانية التعديل وإدارة الجلسات)، وبروفايل مستخدم آخر (مع أزرار المتابعة/البلوك).
 - الاستكشاف والبحث (Explore / Search): للبحث عن المستخدمين والمنشورات وعرض نتائج ذكية.
 - تفاصيل المنشور (Post Details): صفحة مخصصة لمنشور معين لعرض التعليقات بالكامل والمشاركات.
- لوحة التحكم والإدارة (Admin Dashboard):
 - صفحة خاصة بالمشرفين لمراجعة التقارير (Reports)، وحظر المستخدمين (Ban)، وحذف المحتوى المخالف.

2. مسارات المستخدم (User Flows)

الـ Flow هو "الرحلة" اللي بيمشي فيها المستخدم عشان يحقق هدف معين. بناءً على ملفك، عندك 3 تدفقات رئيسية معقدة لازم ترسمها في بالك قبل الكود:

- تدفق المصادقة وإدارة الجلسات (Auth & Session Flow):
 - المستخدم ببسجل دخول -> بياستلم AccessToken و RefreshToken -> لو الـ Access انتهت صلاحيته، الفرونت أند بيعمل طلب تلقائي (Intercept) بالـ Refresh token عشان يجيب واحد جديد بدون ما يخرج المستخدم. لو عمل Logout من جهاز معين، بيتم إلغاء الجلسة دي بس (Multi-session management).
- تدفق النشر والتفاعل (Content Creation Flow):
 - المستخدم بيضغط "إنشاء بوست" -> بيفتح Modal -> بيختار ميديا (صور/فيديو) -> الفرونت أند بيعمل Upload للميديا وبيظهر Indicator (نسبة التحميل) -> بعد النجاح، البوست بينزل في الـ Feed فوراً (أو باستخدام Optimistic UI لإظهاره قبل حتى ما سيرفر الباك أند يرد، لضمان تجربة سريعة جداً).
- تدفق التفاعل مع القصص (Stories Flow):
 - القصص محتوي مؤقت. الـ Flow هنا بيشبه إنستغرام: تضغط على الدائرة -> يفتح Full-screen overlay -> شريط تقدم (Progress Bar) شغال لكل قصة -> تنقل تلقائي بين القصص أو بالضغط على اليمين/اليسار.

3. إدارة الحالة (State Management)

بما إن السيستم "Enterprise-ready"، تقسيم الـ State هنا حيوي جداً لتفادي بطء التطبيق:

الـ Global State (حالة التطبيق العامة)

حاجات محتاجها في كل مكان في التطبيق:

- **Auth State**: بيانات المستخدم الحالي (User Object) وهل هو مسجل دخول أم لا.
- **UI State**: القوائم الجانبية (Sidebars)، الـ Dark/Light mode، والـ Modals المفتوحة.
- **Notifications State**: عدد الإشعارات غير المقروءة والتنبيهات اللحظية.

الـ Server Cache State (بيانات السيرفر)

البيانات القادمة من الـ API (يفضل إدارتها بأدوات مثل React Query أو RTK Query):

- قائمة البوستات في الـ Feed، بيانات التعليقات، نتائج البحث، وقائمة المتابعين. (هنا نحتاج تطبيق الـ Infinite Scroll أو الـ Pagination عشان متجيبش كل البيانات مرة واحدة).

4. حماية المسارات (Route Guards)

الـ Guards هي بوابات الأمان اللي بتحدد مين يشوف إيه. بمساعدة الـ React Router أو الـ Next.js Middleware، هتقسم مساراتك كالتالي:

نوع المسار (Route Type)	الصفحات المتاحة تحت هذا المسار	شرط الوصول (Guard Condition)
Guest Routes	Login, Register, Forgot Password	فقط المستخدمين غير المسجلين. (لو مسجل، بيحصل له Redirect للـ Feed).
Protected Routes	Feed, Profile, Settings, Create Post	فقط المستخدمين المسجلين (لديهم Token صلاحيته سارية).
Admin Routes	Dashboard, Reports, Moderation Panel	فقط لو كان الـ role الخاص بالمستخدم في الـ State يساوي Admin.

5. هيكل التنقل (Navigation Structure)

لتوفير تجربة مستخدم مريحة، الهيكل هيتوزع حسب نوع الجهاز (Responsive Navigation):

- **على الشاشات الكبيرة (Desktop):** شريط جانبي ثابت (Sidebar) يحتوي على: الرئيسية، استكشاف، الإشعارات، الرسائل (مستقبلاً)، الملف الشخصي، وزر إنشاء بوست سريع.
- **على الموبايل (Mobile UX):** شريط سفلي (Bottom Navigation Bar) للأزرار الأكثر استخداماً (Home, Search, Create, Notifications, Profile).

6. أولويات الميزات (Features Priority / MVP)

عشان منتشتش، قسم الشغل على مراحل (Sprints). بما إن الباك آند جاهز، ابدأ بـ المنتج الأدنى القابل للإطلاق (MVP) ثم الميزات المتقدمة:

- **P0 (الأولوية القصوى - الأساس):**
 - نظام الـ Auth الكامل (Login/Register/Tokens).
 - الـ Feed (عرض البوستات) والـ Profile الشخصي.
 - عمليات الـ CRUD الأساسية للبوستات (إنشاء، تعديل، حذف، رفع ميديا).
- **P1 (أولوية متوسطة - التفاعل الاجتماعي):**
 - نظام المتابعة (Follow/Unfollow) والبلوك لتشكيل الـ Social Graph.
 - التفاعلات (Likes, Comments, Saves).
 - البحث (Search) عن المستخدمين والبوستات.
- **P2 (ميزات متقدمة ومكملة):**
 - الـ Stories (القصص المؤقتة وتفاعلاتها).
 - لوحة تحكم الأدمن ونظام التقارير (Reports).
 - نظام الإشعارات (Notifications).

Phase 2 :

FRS (Functional Requirements Specification)

Document Information

Value	Item
Social Media Platform	Product
1.0	Version
Functional Requirements Specification	Type
Frontend + Backend	Scope
MERN Stack	Architectur
Planning Phase	Status

Functional Modules

Module 1 — Authentication & Account Management

Purpose

.Manage user identity, authentication, session lifecycle, and account recovery

Features

Registration

:User shall be able to

- .Create new account •
 - .Verify email •
 - .Set password •
 - .Receive validation feedback •
-

Login

:User shall be able to

- .Login using email/password •
 - .Receive access token •
 - .Create refresh session •
-

Logout

:User shall be able to

- .Logout current session •
 - .Logout all sessions •
-

Password Reset

:User shall be able to

- .Request reset link •

- .Set new password
 - .Invalidate previous tokens
-

Email Verification

:User shall be able to

- .Verify ownership of email
 - .Resend verification email
-

Module 2 — Profile Management

Features

:User shall be able to

- .Update profile image
 - .Update cover image
 - .Update bio
 - .Update display name
 - .Manage privacy settings
-

Privacy Modes

Public
Followers Only
Private

Module 3 — Social Graph

Features

:User shall be able to

- .Follow user
- .Unfollow user
- .View followers
- .View following
- .Block user

.Unblock user •

Module 4 — Posts

Features

:User shall be able to

- .Create post •
 - .Edit post •
 - .Delete post •
 - .View feed •
 - .View profile posts •
-

Media Support

Image
Video

Visibility

Public
Followers Only
Private

Module 5 — Comments

Features

:User shall be able to

- .Create comment •
 - .Edit comment •
 - .Delete comment •
 - .View comments •
-

Module 6 — Reactions

Features

:User shall be able to

- .Like post
- .Unlike post

Module 7 — Saved Posts

Features

:User shall be able to

- .Save post
- .Unsave post
- .View saved collection

Module 8 — Stories

Features

:User shall be able to

- .Upload story
- .Delete story
- .View stories
- .View story viewers

Supported Types

Image
Video

Lifecycle

Auto expire after 24 hours

Module 9 — Notifications

Notification Types

Follow Notification

:Generated when

User follows another user.

Like Notification

:Generated when

User likes a post.

Comment Notification

:Generated when

User comments on a post.

Mention Notification

:Generated when

User is mentioned.

Story Notification

:Generated when

Story interaction occurs.

Module 10 — Search

Search Targets

Users
Posts

Search Capabilities

- .Partial match
 - .Keyword match
 - .Pagination
 - .Sorting
-

Module 11 — Reports

Report Targets

User
Post
Comment

User Actions

:User shall be able to

- .Submit report
 - .Select reason
 - .Add optional description
-

Module 12 — Administration

User Moderation

:Admin shall be able to

- .View users
 - .Ban users
 - .Delete users
-

Ban Rules

:When account is banned

- .Access is denied
- .Login blocked
- .Registration using same email prohibited

Content Moderation

:Admin shall be able to

- .Delete posts
 - .Delete comments
 - .Review reports
-

Business Rules

BR-01

.Email must be unique

BR-02

.Username must be unique

BR-03

.Only authenticated users may create content

BR-04

.Only owners may edit their content

BR-05

.Admins may override ownership restrictions

BR-06

:Blocked users cannot

- .Follow each other
 - .View private content
 - .Interact with content
-

BR-07

.Deleted content cannot be restored by users

BR-08

.Stories automatically expire after 24 hours

BR-09

.Banned users cannot access platform resources

BR-10

.Users cannot like the same post twice

BR-11

.Users cannot follow themselves

BR-12

.Private accounts require follow approval

BR-13

.Saved posts are private to account owner

BR-14

.Notifications belong only to recipient

BR-15

.Reports are visible only to administrators

Acceptance Criteria

Registration

Success

:Given

Valid user information

:When

User submits registration form

:Then

Account created successfully
Verification email sent

Login

:Given

Valid credentials	
	:When
User submits login form	
	:Then
Access token issued	
Session created	
User redirected	

Create Post

	:Given
Authenticated user	
	:When
User submits post	
	:Then
Post created	
Feed updated	

Follow User

	:Given
Two valid users	
	:When
User clicks follow	
	:Then
Relationship created	
Notification generated	

Save Post

:Given

Existing post

:When

User saves post

:Then

Post appears in saved collection

Report Content

:Given

Existing content

:When

User submits report

:Then

Report stored

Admin can review report

(MVP Scope (Recommended

MVP

Authentication

Profiles

Follow System

Posts

Comments

Likes

Saved Posts

Search

Notifications

Admin Panel

V2

Stories
Reporting System
Blocking System
Advanced Moderation

V3

Realtime Notifications
Realtime Presence
Recommendation Engine

Phase 3 :

(Software Requirements Specification (SRS #

Project Information ##

Item		Value
Project		Social Media Platform
Version		1.0
Architecture		MERN Stack
Frontend	React + Vite + TypeScript	
Backend	Node.js + Express + MongoDB	
Purpose	Production-Oriented Portfolio Project	
Status	Planning Phase	

System Overview .1 #

The Social Media Platform is a web-based social networking application that enables users to create content, interact with other users, discover content, manage personal profiles, and participate in a secure social ecosystem

The platform is designed with scalability, maintainability, security, and extensibility as primary engineering goals

The system follows a modular architecture to support future feature expansion without requiring major structural changes

System Objectives .2

:The system shall

- .Provide secure authentication *
- .Support user-generated content *
- .Support social interactions *
- .Support profile privacy controls *
- .Support content moderation *
- .Support scalable frontend architecture *
- .Support future realtime integration *
- .Support long-term maintainability *

Stakeholders .3

Primary Users

Platform Users *
Content Creators *
Community Members *

Secondary Users

Administrators *
Future Moderators *

Technical Stakeholders

Frontend Developer *
Backend Developer *
Future Engineering Teams *

Non-Functional Requirements .4

NFR-01 Performance

Feed Loading

The system should load feed data within acceptable response times under normal .operating conditions

:Target

Initial request < 2 seconds *

Cached request < 500ms *

Pagination ###

:The platform shall implement pagination for

Feed *

User posts *

Notifications *

Search results *

Comments *

:Purpose

Reduce payload size *

Improve rendering performance *

Reduce memory consumption *

Media Optimization ###

:Images and videos should

Be validated before upload *

Use lazy loading *

Prevent unnecessary bandwidth consumption *

NFR-02 Scalability ##

:The architecture shall support future migration toward

Redis *

WebSockets *
CDN storage *
Object storage *
Microservices *

.without requiring major application redesign

Modular Growth ###

:New features shall be introduced through

Feature modules *
Shared abstractions *
Domain entities *

.without modifying unrelated modules

NFR-03 Maintainability ##

:The codebase shall follow

SOLID principles *
Feature-based architecture *
Separation of concerns *
Reusable abstractions *

Code Organization ###

:Frontend structure shall maintain

txt``
app
features
entities
shared
widgets
pages
processes

...

.as independent architectural layers

NFR-04 Security ##

Authentication Security ###

:The system shall

Use JWT authentication *

Use refresh sessions *

Validate tokens *

Protect private routes *

Authorization Security ###

:The system shall

Enforce role validation *

Enforce ownership validation *

Protect administrative resources *

Input Validation ###

.All user input shall be validated

:Examples

Registration *

Login *

Comments *

Posts *

Reports *

Upload Security ###

:Uploaded media shall be validated using

- MIME type validation *
- File size validation *
- File extension validation *
- Magic bytes validation *

Session Security

:The platform shall support

- Session revocation *
- Multi-device sessions *
- Session expiration *

NFR-05 Reliability

:The platform shall

- .Prevent application crashes from invalid requests *
- .Handle API failures gracefully *
- .Display meaningful error messages *
- .Recover from temporary failures when possible *

NFR-06 Availability

:The application should remain operational during

- High interaction activity *
- Large feed browsing sessions *
- Heavy notification usage *

.without significant degradation

NFR-07 Accessibility

:The platform should support

Keyboard navigation *

Screen readers *

Proper form labeling *

Semantic HTML *

.Future improvements may expand accessibility compliance

NFR-08 Usability

:The user experience shall prioritize

Simplicity *

Discoverability *

Consistency *

Mobile responsiveness *

NFR-09 Compatibility

:Frontend shall support

Browsers ###

Chrome *

Edge *

Firefox *

Safari *

.Recent stable versions only

Devices ###

Desktop *

Tablet *

Mobile *

NFR-10 Observability

:The architecture should support future integration with

Logging systems *

Monitoring systems *

Analytics systems *

Error tracking systems *

:Examples

Sentry *

Datadog *

ELK Stack *

System Constraints .5

Technical Constraints

C-01 Frontend Stack

:Frontend must use

React *

Vite *

TypeScript *

React Router *

Tailwind CSS *

shadcn/ui *

Zustand *

TanStack Query *

Axios *

React Hook Form *

Zod *

C-02 Backend Stack

:Backend must use

Node.js *

Express *
MongoDB *
Mongoose *

C-03 Authentication Model

.Authentication shall remain JWT-based

C-04 Database

.Primary database shall remain MongoDB

C-05 Repository Structure

.Frontend and backend must remain separated repositories

C-06 Architecture Constraint

.Feature modules must not directly depend on other feature modules

:Allowed dependencies

txt``

Feature → Shared

Feature → Entity

:Not Allowed

Feature → Feature

...

C-07 Privacy Constraint

.Private accounts must not expose restricted content to unauthorized users

C-08 Content Ownership Constraint

.Users may only modify content they own unless administrator permissions apply

Assumptions .6

A-01 Internet Access

.Users are assumed to have internet connectivity

A-02 Email Delivery

:Email service is assumed to successfully deliver

Verification emails *

Password reset emails *

A-03 Browser Support

.Users are assumed to use modern browsers

A-04 Media Storage

.Uploaded media is assumed to be stored successfully by the media subsystem

A-05 Authentication Provider

.JWT infrastructure remains available and operational

A-06 User Behavior

.Users are expected to follow platform rules and acceptable use policies

A-07 Future Expansion

:Future versions may introduce

Realtime notifications *

Direct messaging *

Groups *

Communities *

Recommendation systems *

Advanced moderation *

.Current architecture should remain compatible with these future additions

Risks .7

R-01 Media Growth

.Media uploads may significantly increase storage requirements

:Mitigation

CDN integration *

External object storage *

R-02 Feed Complexity

.Feed generation may become expensive as user count grows

:Mitigation

Pagination *

Caching *

Query optimization *

R-03 Notification Volume

.High engagement may generate excessive notifications

:Mitigation

Notification batching *

Future realtime optimization *

R-04 Feature Expansion

.Future feature growth may increase complexity

:Mitigation

Feature isolation *

Strict architectural boundaries *

Modular design *

SRS Completion Criteria

:The system specification will be considered satisfied when

.Functional requirements are implemented *

.Security requirements are enforced *

.Architectural boundaries are respected *

.Performance requirements are met *

.Future scalability remains achievable *

.The platform supports planned V2 and V3 growth without structural redesign *

المرحلة الرابعة هي التي ستحول كل ما كتبناه سابقًا إلى مخطط هندسي فعلي. لكن بصراحة، **System Design + Database Design + API Design + Security Design + Frontend Design** لمشروع بهذا الحجم لن يمكن تقديمها بجودة عالية في رد واحد دون أن تصبح مختصرة بشكل مغل. لهذا أوصي بالتقسيم التالي:

Phase 4.1 → System Design

Phase 4.2 → Database Design

Phase 4.3 → API Design

Phase 4.4 → Security Design

Phase 4.5 → Frontend Design

Phase 4.1 — System Design

System Design

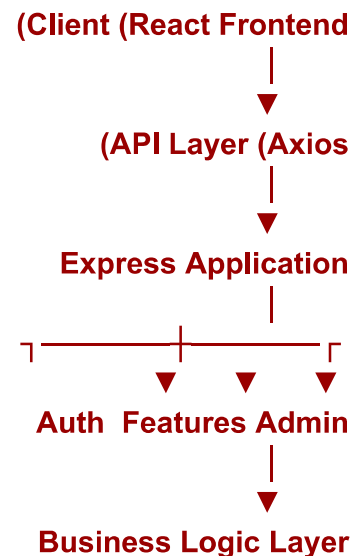
System Overview

The Social Media Platform is designed as a layered MERN application composed of independent modules that communicate through clearly defined boundaries.

The architecture prioritizes

- Scalability
- Maintainability
- Security
- Separation of Concerns
- Future Feature Expansion

High Level Architecture





Architectural Layers

Frontend Layer

:Responsibilities

- Rendering UI •
- State Management •
- API Communication •
- Form Handling •
- Route Protection •

:Technology

- React •
- TypeScript •
- React Router •
- Zustand •
- TanStack Query •

API Layer

:Responsibilities

- Request Handling •
- Authentication Validation •
- Authorization •
- Validation •
- Rate Limiting •

:Technology

- Express.js •
-

Domain Layer

:Responsibilities

- Business Logic •
- Ownership Rules •
- Social Relationships •
- Notification Creation •

:Examples

- Follow User •
 - Create Post •
 - Save Post •
 - Report Content •
-

Persistence Layer

:Responsibilities

- Data Storage •
- Query Execution •
- Data Relationships •

:Technology

- MongoDB •
 - Mongoose •
-

Major Modules

Authentication Module

:Handles

- Registration •
- Login •
- Logout •
- Email Verification •
- Password Reset •

Session Management •

User Module

:Handles

- Profiles** •
 - Privacy Settings** •
 - Following** •
 - Followers** •
 - Blocking** •
-

Post Module

:Handles

- Create Post** •
 - Edit Post** •
 - Delete Post** •
 - Feed Generation** •
 - Post Visibility** •
-

Comment Module

:Handles

- Comment Creation** •
 - Comment Updates** •
 - Comment Removal** •
-

Story Module

:Handles

- Story Creation** •
- Story Viewing** •

Story Expiration •

Notification Module

:Handles

- Like Notifications •
 - Follow Notifications •
 - Mention Notifications •
 - Comment Notifications •
 - Story Notifications •
-

Report Module

:Handles

- User Reports •
 - Post Reports •
 - Comment Reports •
-

Admin Module

:Handles

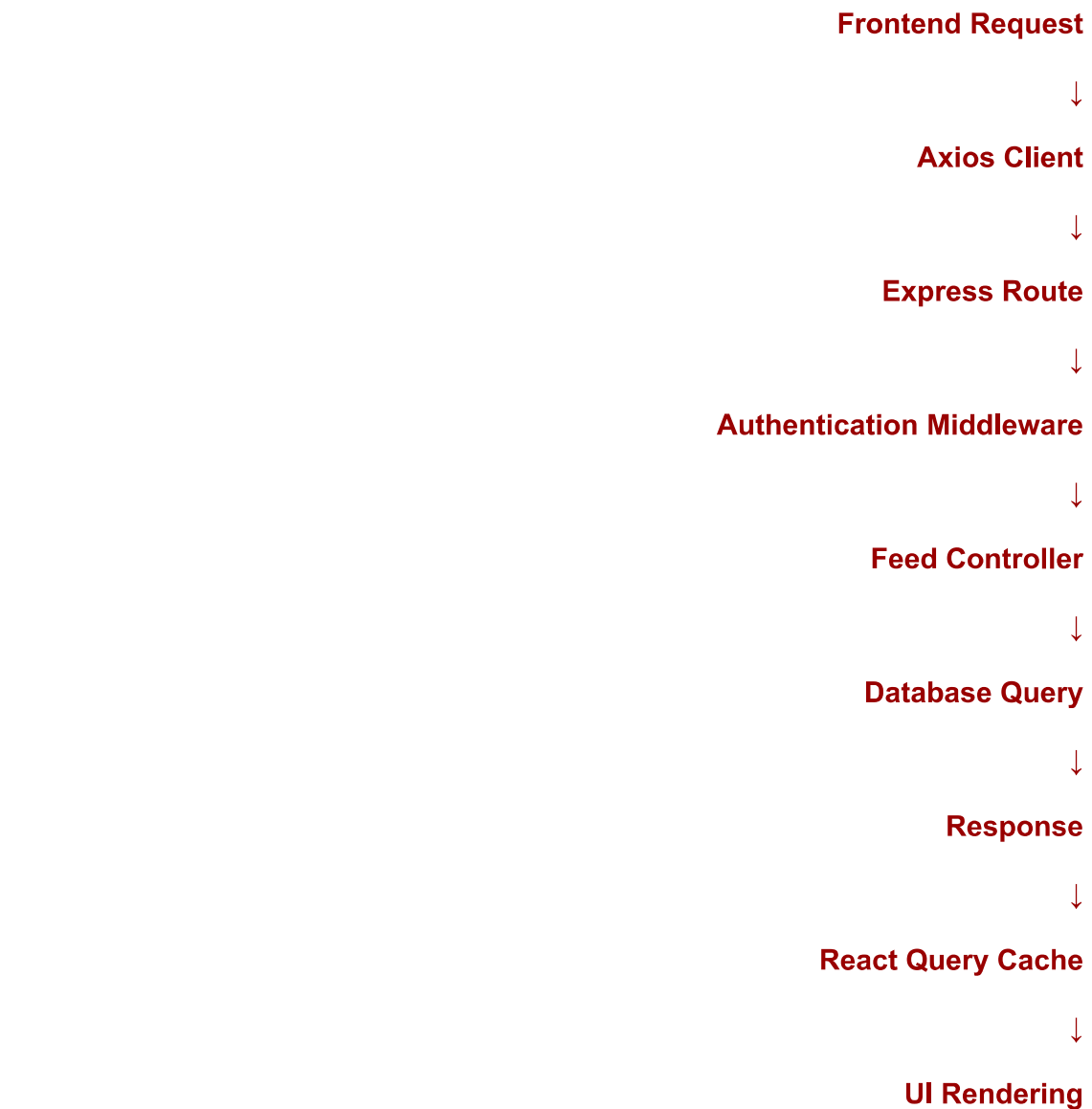
- User Moderation •
 - Content Moderation •
 - Report Management •
-

Data Flow Example

Feed Retrieval

User Opens Feed





Scalability Strategy



CDN
WebSockets
Object Storage
Microservices

Caching Strategy

:Current

In-memory cache

:Future

Redis Cache

:Targets

- Feed •
 - User Profiles •
 - Public Posts •
-

Deployment Vision

:Frontend

Vercel •

:Backend

Render / Railway / VPS •

:Database

MongoDB Atlas •

:Media

Cloudinary / S3 •

System Design Principles

- SOLID
- DRY
- KISS
- Feature Isolation
- Domain Separation
- Secure By Default
- Scalable By Design